

Program5 :Demonstrate CI Integration by running jenkins in a docker container

Folder structure

```
|  
|--Dockerfile  
|--docker-compose.yaml
```

Step1: create following files

Dockerfile

```
FROM jenkins/jenkins:lts  
  
#switch to root users to install tools  
  
USER root  
#Install docker inside jenkins container  
  
RUN apt-get update && apt-get install -y docker.io  
  
#switch back to jenkins user  
USER jenkins
```

docker-compose.yaml

```
version: '3.8'  
services:  
  jenkins:  
    build: .  
    container_name: jenkins-ci  
    ports:  
      - "8080:8080"  
      - "50000:50000"  
    volumes:  
      - jenkins_home: /var/jenkins_home  
      - /var/run/docker.sock: /var/run/docker.sock  
volumes:  
  jenkins_home:
```

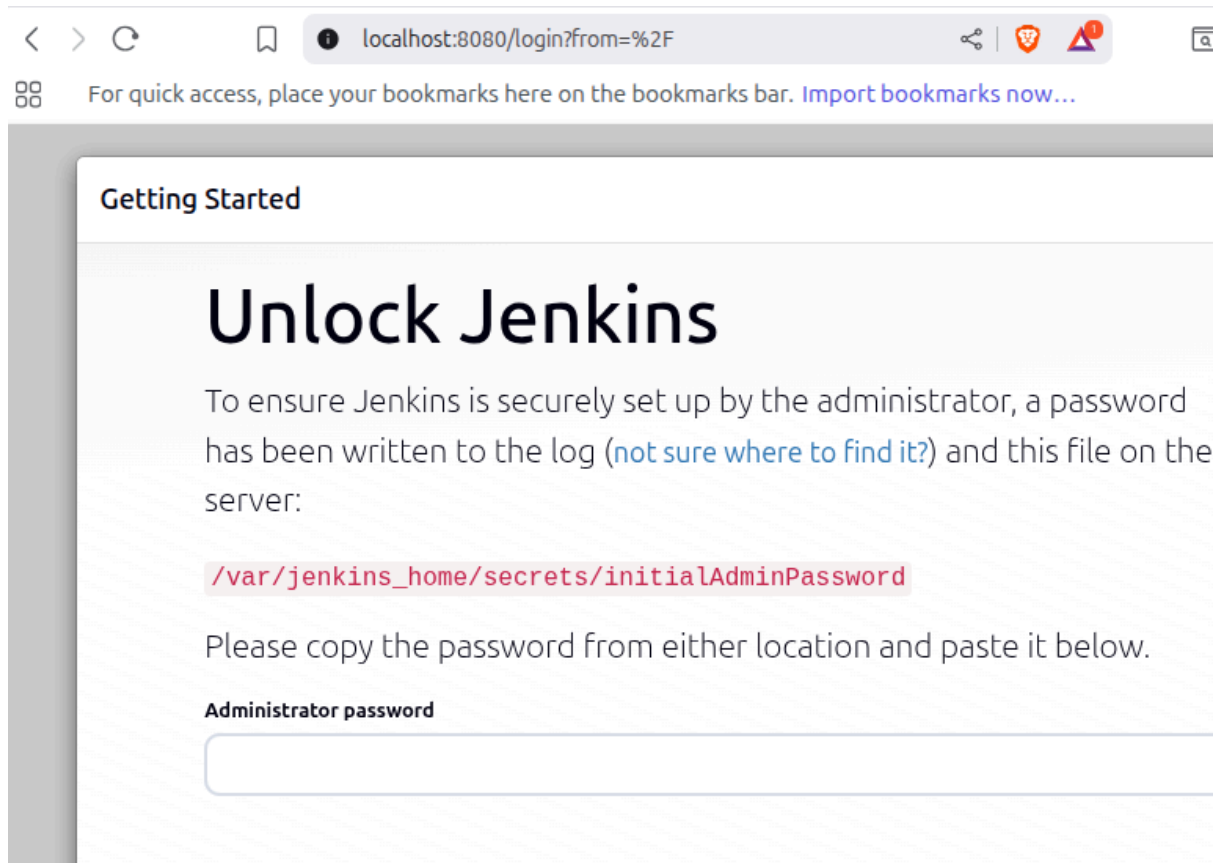
Step2:Build and run the containers

Using Dockerfile

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/program5$ docker build -t program5 .
[+] Building 487.3s (8/8) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 254B
=> [internal] load metadata for docker.io/jenkins/jenkins:lts
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b
=> => resolve docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b
=> => sha256:b768bbb699b5f1b6579e23e7253ae1157a339a20a2e9befdf8e03923e 2.58kB / 2.58kB 0.0s
=> => sha256:75109de20007f9345278f298f4534d443409e234ba81225815caa81 12.71kB / 12.71kB 0.0s
=> => sha256:13cc39f8244ac66bf1dd9149e1da421ab1bbc80d612dc14fe3687 49.29MB / 49.29MB 152.6s
=> => sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13 3.12kB / 3.12kB 0.0s
=> => sha256:5687d4df0728e3558596e7d718163b2e812a00cf5dd2624459c33 65.08MB / 65.08MB 330.8s
=> => sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127 5.24MB / 5.24MB 12.9s
=> => sha256:68cdef48009293f6a12f5c50b91fcfe4194bd6293334d8a31a45b1ed 1.23kB / 1.23kB 13.2s
=> => sha256:e0e73f34dee93322ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c3 183B / 183B 23.5s
=> => sha256:687044b042c5d6d57e2ce207c000ee159f2382cb84cb8d80688c35 95.32MB / 95.32MB 93.0s
=> => sha256:501deb283b89e30fd3f1f4e27a635bcc1739485e8ba805034552e3d4eaac 189B / 189B 93.9s
=> => sha256:a9230f7c9c31780c6b1dc4f2aa7e1073df2fdf88e9c009f6874e7221 6.29MB / 6.29MB 97.8s
=> => sha256:d7dd2e63023897146ec4b913a460382968fbbd1cc3682e36f6507 72.05MB / 72.05MB 137.9s
=> => sha256:3fb70446889ebcc58e91f43f90aafa314798941c318ce016ea62068 2.23kB / 2.23kB 138.7s
=> => sha256:986be6969b0a04975a2e559b30a96d766bf360d0b340b2789b61168 1.29kB / 1.29kB 139.0s
=> => sha256:d2519f76d4c6527e03084f2742da3ae15b0c5097d464be4dd93de4f5737 392B / 392B 140.7s
=> => extracting sha256:13cc39f8244ac66bf1dd9149e1da421ab1bbc80d612dc14fe368753e7be17b 1.8s
=> => extracting sha256:5687d4df0728e3558596e7d718163b2e812a00cf5dd2624459c3315dce34b3 2.5s
=> => extracting sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127c01214 0.1s
=> => extracting sha256:68cdef48009293f6a12f5c50b91fcfe4194bd6293334d8a31a45b1ed63cb91 0.0s
=> => extracting sha256:e0e73f34dee93322ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c311 0.0s
=> => extracting sha256:687044b042c5d6d57e2ce207c000ee159f2382cb84cb8d80688c35e7f1ed86 0.4s
=> => extracting sha256:501deb283b89e30fd3f1f4e27a635bcc1739485e8ba805034552e3d4eaac0b 0.0s
=> => extracting sha256:a9230f7c9c31780c6b1dc4f2aa7e1073df2fdf88e9c009f6874e7221acbb62 0.1s
=> => extracting sha256:d7dd2e63023897146ec4b913a460382968fbbd1cc3682e36f650727a1bfb6d8 0.9s
=> => extracting sha256:3fb70446889ebcc58e91f43f90aafa314798941c318ce016ea62068a3de539 0.0s
=> => extracting sha256:986be6969b0a04975a2e559b30a96d766bf360d0b340b2789b611688ba8bbe 0.0s
=> => extracting sha256:d2519f76d4c6527e03084f2742da3ae15b0c5097d464be4dd93de4f5737493 0.0s
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io
=> [2/2] RUN apt-get update && apt-get install -y docker.io
=> exporting to image
=> => exporting layers
=> => writing image sha256:bfa7281fd3ad61ddc63b7e92e37dca217dc2caeb16dffdf1f6b83e40f832 145.5s
=> => naming to docker.io/library/program5 1.7s
=> => 1.7s
=> => 0.0s
=> => 0.0s
```

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/program5$ docker run -p 8080:8080 -p 50000:50000 --name jenkins_ci program5
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2025-12-26 05:02:58.522+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extract
ion from war file
2025-12-26 05:02:59.855+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath
: Empty contextPath
2025-12-26 05:02:59.929+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty
-12.0.25; built: 2025-08-11T23:52:37.219Z; git: a862b76d8372e24205765182d9ae1d1d333ce2ea; jvm
21.0.9+10-LTS
2025-12-26 05:03:00.471+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitSer
vlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2025-12-26 05:03:00.550+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Sessi
on workerName=node0
2025-12-26 05:03:01.054+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins
home directory: /var/jenkins_home found at: EnvVars.masterEnvVars.get("JENKINS_HOME")
2025-12-26 05:03:01.212+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Starte
d oeje9n.ContextHandler$CoreContextHandler@3c782d8e{Jenkins v2.528.3/,b=file:///var/jenkins_h
ome/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@572e6fd9{S
TARTED}}
2025-12-26 05:03:01.229+0000 [id=1] INFO o.e.i.server.AbstractConnector#doStart: Starte
```

Output:



Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

/var/jenkins_home/secrets/initialAdminPassword

Please copy the password from either location and paste it below.

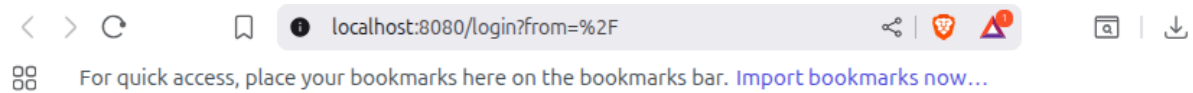
Administrator password

Step3: Using Docker compose

Docker compose up

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/program5$ docker compose up
WARN[0000] /home/1rv24mc055_kishan_s/lab/program5/docker-compose.yaml: the attribute `version`
is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 2.9s (9/9) FINISHED
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 535B                                           0.0s
=> [internal] load build definition from Dockerfile                     0.0s
=> => transferring dockerfile: 254B                                     0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lts         2.7s
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io         0.0s
=> [internal] load .dockerignore                                         0.0s
=> => transferring context: 2B                                           0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2b 0.0s
=> CACHED [2/2] RUN apt-get update && apt-get install -y docker.io    0.0s
=> exporting to image                                                  0.0s
=> => exporting layers                                                  0.0s
=> => writing image sha256:0ef795b3fcdcd77ed056b56c2a388742809eb5665e61fd4fad4a94029709 0.0s
=> => naming to docker.io/library/program5-jenkins                    0.0s
=> resolving provenance for metadata file                               0.0s
[+] Running 4/4
✔ program5-jenkins              Built              0.0s
✔ Network program5_default      Created        0.1s
✔ Volume program5_jenkins_home  Created        0.0s
✔ Container jenkins-ci          Created        0.0s
Attaching to jenkins-ci
jenkins-ci | Running from: /usr/share/jenkins/jenkins.war
jenkins-ci | webroot: /var/jenkins_home/war
jenkins-ci | 2025-12-26 05:08:38.879+0000 [id=1] INFO winstone.Logger#logInternal: B
eginning extraction from war file
jenkins-ci | 2025-12-26 05:08:40.174+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandle
r#setContextPath: Empty contextPath
jenkins-ci | 2025-12-26 05:08:40.245+0000 [id=1] INFO org.eclipse.jetty.server.Serve
r#doStart: jetty-12.0.25; built: 2025-08-11T23:52:37.219Z; git: a862b76d8372e24205765182d9aeld
1d333ce2ea; jvm 21.0.9+10-LTS
jenkins-ci | 2025-12-26 05:08:40.807+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorPr
ocessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspSer
vlet
jenkins-ci | 2025-12-26 05:08:40.901+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManage
r#doStart: Session workerName=node0
```

Go to browser and type <http://localhost:8080>



Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

`/var/jenkins_home/secrets/initialAdminPassword`

Please copy the password from either location and paste it below.

Administrator password